

To V or not to V

A medical tradition that saved the life of billions is suddenly being called into question. Here's the final word.

Space age: Europa

We're actually making our way to Europa as we speak. Hopefully, there's no perfectly black cuboid awaiting us.

The Sci-Tech Leaderboard

A leaderboard for the extreme in scientific discoveries, achievements and inventions.

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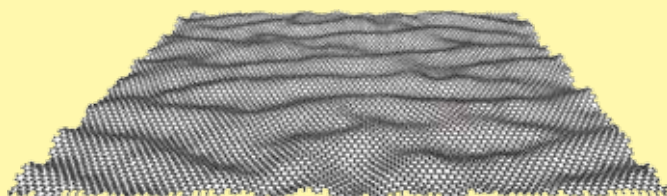
Every once in a while we humans do something right. Something perfect and pristine that deserves to be applauded. In the world of science many of these achievements and

discoveries warrant universal acclaim, such as a Nobel Prize. In other cases they find their way onto a list with superlatives in the title. This is one of those lists. So join us as we take a survey of the biggest, baddest and awesomest of things that we humans made. Until some future time when they all will be obsolete, replaced by even newer and more awesome achievements and discoveries.

The Roundest Object Ever Made

Why does this matter? Interestingly enough, the existence of the perfect sphere lies at the core of our measurement system - specifically the "kilogram problem". We use this standard unit of measurement on a daily basis but what does it actually mean? Well, the kilogram is the only unit of measurement that is tied to a specific physical object. That object is known as the international prototype kilogram or IPK and is a fabricated of a platinum-iridium alloy kept securely at the International Bureau of Weights and Measurements (BIMP). Numerous copies are in other nations but due to its elemental nature it is not a constant.

To solve the consistency problem of this standard the Avogadro Project was started which sought to measure and define kilograms as a specific number of silicon atoms. The manifestation of this design was accomplished by the Australian Center for Precision Optics and engineer Achim Leistner in 2008. The sphere was created using high precision tools to achieve its 93.75 millimeters of diameter accurate to within 0.3 nanometers. This makes this sphere makes it the roundest object ever made. The BIMP has however decided to use the Planck constant to define the base line of a kilogram.



The Flattest Material in the World

In 2010 the Nobel Prize for Physics was awarded to Russian scientists Andre Geim and Konstantin Novoselov at the University of Manchester in the United Kingdom. They were credited with the discovery of a carbon based material called graphene which is the flattest material in the world. And when we say "flattest" we don't mean really, really thin - we are talking about the width of an atom thin - it's just that flat. Graphene manifests as a sheet-like, near two dimensional substance, making it practically invisible. In addition to its uber cool qualities, we also notice that graphene is 100 times stronger than steel thanks to its hexagonally patterned carbon atom bonds. Its potential applications as an electric conductor is remarkable and because it's cheap to produce, it's ideal for use in computers and micro-circuitry, which make it a prime candidate for adoption by tech industries. It's worth noting that the Russian duo achieved this using nothing but scotch tape and granite. On a side note, one of them also won an Ig Nobel for levitating frogs using magnetism. Just thought we'd throw that in there and hey, if we had our way, he'd win a Nobel prize for the frogs. It's not a sight we see every day.